The F = 1 target-in-walk arm at k = 28 and 32, single-target ecbench (2026-11-08)

Class. Measured, single-target ecbench sessions under a protocol frozen before data (PROTOCOL.md hashes in `data/frozen.sha256`, freeze commit `e6a759cc72`). Counted units only (ecbench GAE, S = cost/sqrt(r)); every comparison at w = 0, 0.1 and 1 under the committed band and under `ecbench band --price-table-work`. Public synthetic Curve25519-family miniatures; no security claim. No pcc claim has been filed; every number is an unverified assertion until it is reviewed [pcc:unverified]. No code change; binary `/home/user/ctarget-f1/release/ecbench` (SHA-256 `d23fab99...`, `data/binary-k{28,32}.sha256`).

Question. The target-in-walk arm had been measured at k = 20 and 24 only. Does its distance to rho-neg hold near 1.4x at k = 28 and 32, where rho-neg's set-up overhead vanishes and both arms sit near their walk floors, and do 32 or 64 jumps pay there, where the set-up they cost is 0.01 sqrt(r) or less?

Answer. The distance holds. The 16-jump arm costs 1.395x [1.345, 1.446] rho-neg at k = 28 and 1.366x [1.267, 1.473] at k = 32 at w = 0, and its lower end stays above 1 at every weight once table inserts are priced like rho's canonicalisations [pcc:unverified]. Against the table-then-probe F = 1 arm it costs 0.906x [0.877, 0.935] and 0.857x [0.811, 0.905] [pcc:unverified]. Its mean S at k = 32 is 1.387 with a walk of 1.368 sqrt(r), 3 % above the second-collision floor of 1.329; rho-neg's search is 0.999 sqrt(r) there, 13 % above its first-collision floor of 0.886, which is why the measured ratio is 1.37x and not the model's 1.5x [pcc:unverified]. More jumps do not pay: 32 jumps give 0.997x [0.954, 1.041] at k = 28 and 0.985x [0.930, 1.044] at k = 32, 64 jumps 0.998x and 0.990x, all intervals containing 1 [pcc:unverified]. Five of seven frozen predictions held; the only failures are the risky P4b (32 jumps strictly below 16 at k = 32) and nothing else [pcc:unverified].

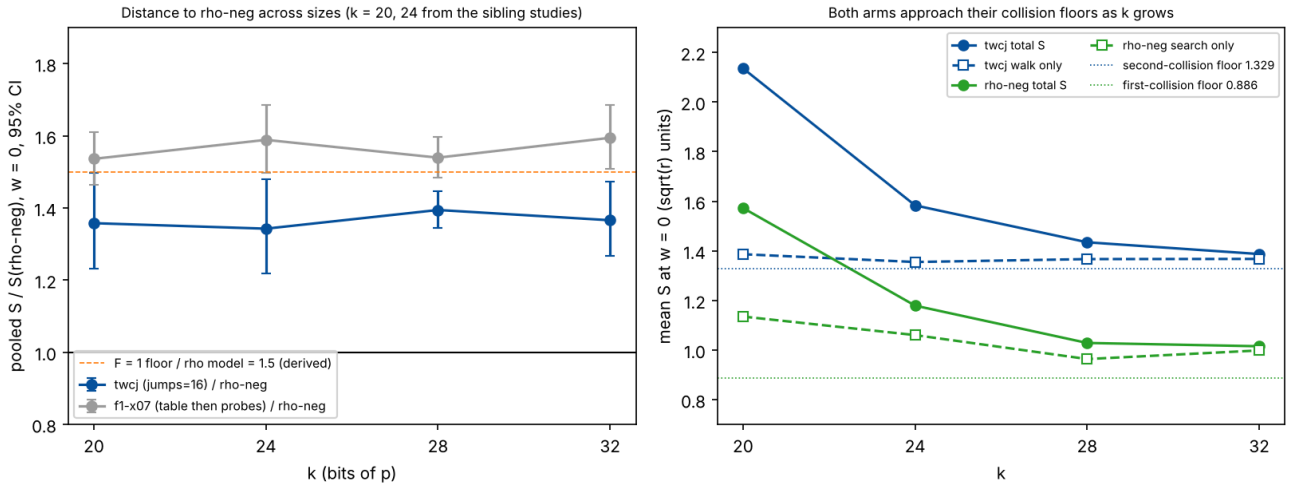


Figure 1: Scaling of the distance to rho

1. Design (as frozen)

8 sessions per size, `targets_per_curve = 1`, 64 measured rounds plus 1 warm-up per arm, `order = alternate`; arms `rho-neg`, `f1-x07` (baseline, `wwalk:x=0.7,g=1,grow=2,seed=run`), `twcj-j16` (primary), `twcj-j32`, `twcj-j64` (`wwalk:x=1.5,g=1,grow=2,tw=1,cj=1,jumps=J,seed=run`), all pipeline arms on `prime-projected-abscissa:size=1,scan_cap=32` with `targets=stride-generator`. Curves: the family miniatures of the autolab holdouts at k = 28 (`r = 33,557,809`) and k = 32 (`r = 536,884,639`). Audit: `ecbench verify --replay 1` per session. Pooling as in the sibling studies. Runtime: 50 s at k = 28, 76 s at k = 32 (`data/run-*.log`).

2. Results (measured)

All 16 sessions completed with 325 of 325 records verified each; every audit reproduced its replay with 0 problems (`data/verify-k*-s*.txt`) [pcc:unverified]. Pooled figures use the 2,560 measured records per size (`data/pooled_measured.json`).

Mean S per arm (committed band; then with table inserts also priced at w):

k	arm	S w=0	S w=0.1	S w=1	w=0.1, inserts priced	w=1, inserts priced
28	rho-neg	1.029	1.125	1.993	1.125	1.993
28	f1-x07	1.584	1.639	2.131	1.735	3.100
28	twcj-j16	1.435	1.435	1.438	1.572	2.805
28	twcj-j32	1.430	1.430	1.433	1.565	2.784
28	twcj-j64	1.432	1.433	1.436	1.566	2.765
32	rho-neg	1.015	1.115	2.014	1.115	2.014

32	f1-x07	1.619	1.677	2.201	1.779	3.218
32	twcj-j16	1.387	1.387	1.388	1.524	2.756
32	twcj-j32	1.367	1.367	1.368	1.501	2.711
32	twcj-j64	1.373	1.373	1.374	1.507	2.717

Pooled ratios with 95 % intervals over 8 single-target sessions:

k	ratio	w = 0	w = 0.1 (committed / inserts priced)	w = 1 (committed / inserts priced)
28	twcj-j16 / rho-neg	**1.395** [1.345, 1.446]	1.275 [1.230, 1.322] / 1.397 [1.347, 1.448]	0.722 [0.695, 0.749] * / 1.407 [1.356, 1.461]
28	f1-x07 / rho-neg	1.540 [1.484, 1.598]	1.456 / 1.542	1.070 * / 1.556
28	twcj-j16 / f1-x07	**0.906** [0.877, 0.935]	0.876 / 0.906	0.675 * / 0.905
28	twcj-j32 / twcj-j16	0.997 [0.954, 1.041]	0.997 / 0.996	0.997 / 0.993
28	twcj-j64 / twcj-j16	0.998 [0.947, 1.053]	0.998 / 0.996	0.998 / 0.986
32	twcj-j16 / rho-neg	**1.366** [1.267, 1.473]	1.244 [1.154, 1.341] / 1.367 [1.267, 1.473]	0.689 [0.639, 0.743] * / 1.368 [1.268, 1.476]
32	f1-x07 / rho-neg	1.595 [1.508, 1.687]	1.504 / 1.595	1.093 * / 1.598
32	twcj-j16 / f1-x07	**0.857** [0.811, 0.905]	0.827 / 0.857	0.631 * / 0.856
32	twcj-j32 / twcj-j16	0.985 [0.930, 1.044]	0.985 / 0.985	0.985 / 0.984
32	twcj-j64 / twcj-j16	0.990 [0.934, 1.049]	0.990 / 0.989	0.990 / 0.986

* The committed band prices rho-neg's per-step canonicalisation and not the walk arms' per-step table inserts; ecbench's committed band verdict for rho-neg against every walk arm is "unresolved inside the band" in all 16 sessions (`data/band-k*.txt`), so no rank is admitted under that reading [pcc:unverified]. Under `--price-table-work` (`data/band-tw-k*.txt`) rho-neg is ahead of every arm at every weight [pcc:unverified].

Phases and walk diagnostics (mean over 512 measured runs per arm, units of sqrt(r))

k	arm	fixed	set-up	walk	collision verify	probes	total	walk steps / sqrt(r) (sd)	collisions per run	set-up GAE
28	rho-neg	-	0.059	0.964	0.006	-	1.029			
28	f1-x07	0.012	0.055 (incl. verify)	0.969		0.548	1.584			
28	twcj-j16	0.012	0.032	1.367	0.024	0	1.435	1.37 (0.48)	2.06	186
28	twcj-j32	0.012	0.044	1.351	0.023	0	1.430	1.35 (0.47)	2.03	255
28	twcj-j64	0.012	0.068	1.329	0.023	0	1.432	1.33 (0.49)	2.01	393
32	rho-neg	-	0.015	0.999	0.002	-	1.015			
32	f1-x07	0.004	0.016 (incl. verify)	1.017		0.582	1.619			
32	twcj-j16	0.004	0.009	1.368	0.007	0	1.387	1.37 (0.52)	2.06	212
32	twcj-j32	0.004	0.012	1.344	0.007	0	1.367	1.34 (0.48)	2.04	290
32	twcj-j64	0.004	0.019	1.343	0.007	0	1.373	1.34 (0.49)	2.01	443

Every twcj record at every jump count solved at rank 2 with zero probe hits on the first trial (3,072 of 3,072) [pcc:unverified]. The set-up matches the frozen model (186 against 180 GAE at k = 28 for 16 jumps; 3.5 to 3.8 GAE per extra jump) [pcc:unverified].

3. Predictions (PROTOCOL.md section 5)

id	prediction	outcome	status
P1	twcj-j16/rho-neg point in [1.30, 1.50] at w = 0, lower end > 1 at w = 0 and (inserts priced) at every weight, both sizes	1.395, 1.366; lower ends 1.345 / 1.347 / 1.356 and 1.267 / 1.267 / 1.268	**held** [pcc:unverified]
P2	twcj-j16/f1-x07 upper end < 1, point in [0.82, 0.93], both sizes	0.906 [0.877, 0.935], 0.857 [0.811, 0.905]	**held** [pcc:unverified]
P3	j16 walk steps in [1.30, 1.42] both sizes; mean S at k = 32 in [1.36, 1.50]	1.367, 1.368; 1.387	**held** [pcc:unverified]
P4	j32/j16 point in [0.95, 1.00] both sizes	0.997, 0.985	**held** [pcc:unverified]

P4b	k = 32 j32/j16 upper end < 1	1.044	**failed** [pcc:unverified]
P5	j64/j16 point in [0.95, 1.01] both sizes	0.998, 0.990	**held** [pcc:unverified]
P6	>= 99.5 % of twcj records at rank 2, no probe, first trial	100 % at every jump count	**held** [pcc:unverified]

4. Reading

- **The distance to rho is flat in k.** With the sibling studies' k = 20 and 24 points (1.358 and 1.343 on the same arm, `research/curve25519_tw_jumps_20261108`), the 16-jump arm is 1.34x to 1.40x rho-neg at w = 0 from k = 20 to 32, and the table-then-probe arm 1.54x to 1.60x [pcc:unverified]. Neither moves toward parity with size.
- **Why 1.37x and not 1.5x.** The derived F = 1 floor ratio 1.5 = 1.329 / 0.886 compares the two walks' model floors. Measured, the twcj walk runs 3 % above 1.329 at every size, while rho-neg's search runs 9 % to 13 % above 0.886 (0.964 and 0.999 sqrt(r) at k = 28, 32; its look-ahead and cycle escapes are counted) [pcc:unverified]. The ratio of the measured walks is 1.37 to 1.42, and the remaining set-up and verification (0.03 to 0.07 sqrt(r) at these sizes) add little. A better rho-neg walk (closer to its own floor) would widen the gap; nothing here changes the reference.
- **Jumps.** The walk shortens with more jumps (1.367 to 1.344 to 1.343 sqrt(r) at k = 32), by about the amount the 20- and 24-bit study saw, but the gain of 0.02 sqrt(r) is inside the session noise (interval half-width 0.05) and is not established [pcc:unverified]. Sixteen jumps remains the default; 32 is not worse at k >= 28.

5. Distance to matched rho (rho-neg, same sessions)

k	arm	w = 0	w = 0.1 (committed / inserts priced)	w = 1 (committed / inserts priced)
28	twcj-j16	1.40x [1.35, 1.45]	1.28x / 1.40x	0.72x (artifact) / 1.41x
28	f1-x07	1.54x	1.46x / 1.54x	1.07x / 1.56x
32	twcj-j16	1.37x [1.27, 1.47]	1.24x / 1.37x	0.69x (artifact) / 1.37x
32	twcj-j32	1.35x [1.25, 1.45]	1.23x / 1.35x	0.68x (artifact) / 1.35x
32	f1-x07	1.60x	1.50x / 1.60x	1.09x / 1.60x

6. Scope and limits

As in the sibling studies: the F = 1 target-in-walk arm is one unknown away from rho (P1 := G makes it a rho walk with a stored table), does no decomposition work, is not decomposition-based index calculus, and proposes no leaderboard or `benchmarks/README.md` row; no VALIDATION.json, replay certificate or validator report exists [pcc:unverified]. Single target only. Local L0 counted units, deterministic and replayed; no wall-clock claim. No A/A arm (it would replicate rho-neg exactly). research-kb and ecdb were not reachable; no novelty claim.

Files

`PROTOCOL.md`, `specs/`, `data/frozen.sha256`, `data/frozen-at.txt`, `scripts/` (`model.py`, `make_specs.py`, `run_sessions.sh`, `pooled_compare.py`, `analyse.py`, `make_figure.py`, `render_pdf.py`), `sessions/k*-s*/{session,spec,plan,host}.json` (`records.jsonl` in `data/sessions-k{28,32}-records.tar.gz`, hashed in the benchmark records), `data/` (model, plans, run logs, both band readings, verify reports, `pooled_measured.json`, binary hashes), `figures/scaling.{svg,png}`, `REPORT.pdf`. Benchmark records: `benchmarks/records/20261108T2051Z-curve25519-tw-scaling-k{28,32}.json`; copies in `benchmarks/20261108-curve25519-tw-scaling/`.